A per-paper dispersion model

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When an AI agent is asked to reproduce the same paper several times, its runs do not agree with each other. This note proposes how to measure that disagreement and how to tell what causes it. Nothing here has been estimated yet. It is written for review before a study is registered.

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1 Research question

When repeat runs of one paper disagree about whether its results reproduce, where does the disagreement come from: the paper, the conditions the run had, or chance?

This is not the question of whether a paper reproduces. Each run already answers that, result by result. The question is how far that answer depends on which run gave it. The two come apart: a paper whose results fail in every run shows no disagreement at all, exactly like a paper whose results reproduce in every run. Disagreement is largest when a result reproduces in about half of the runs.

Three questions make that precise:

1. **How large is it?** For each result a paper reports, how likely is it that two independent runs disagree about whether it reproduced?
2. **What does it belong to?** How much of the disagreement comes from the paper and the materials its authors released, how much from the environment of the run (which AI agent ran it, what it could reach on the network, which copy of the paper it was given), and how much is left as chance between runs under the same conditions?
3. **Is it a property of the paper?** Does the amount of disagreement differ reliably from paper to paper, does the ranking of papers hold when the environment changes, and does disagreement rise on exactly the results affected when a stated detail is withheld from a paper?

The answer to the first question, averaged over a paper's results, is the paper's disagreement. It is written H below: the chance that two runs disagree about one result of the paper. It is 0 when every run agrees on every result, and at most 0.5, when each result reproduces in half of the runs.

The answer to the third question decides what an audit may report. If it is yes, H can be reported as a measure of how completely the paper and its materials pin down its own analysis.

If not, the disagreement describes the auditing instrument, and a single run's verdict has to be reported as one draw under stated conditions.

The estimands and the model-free estimator below answer the first question, the model the second, and the three tests the third. The study in one picture:

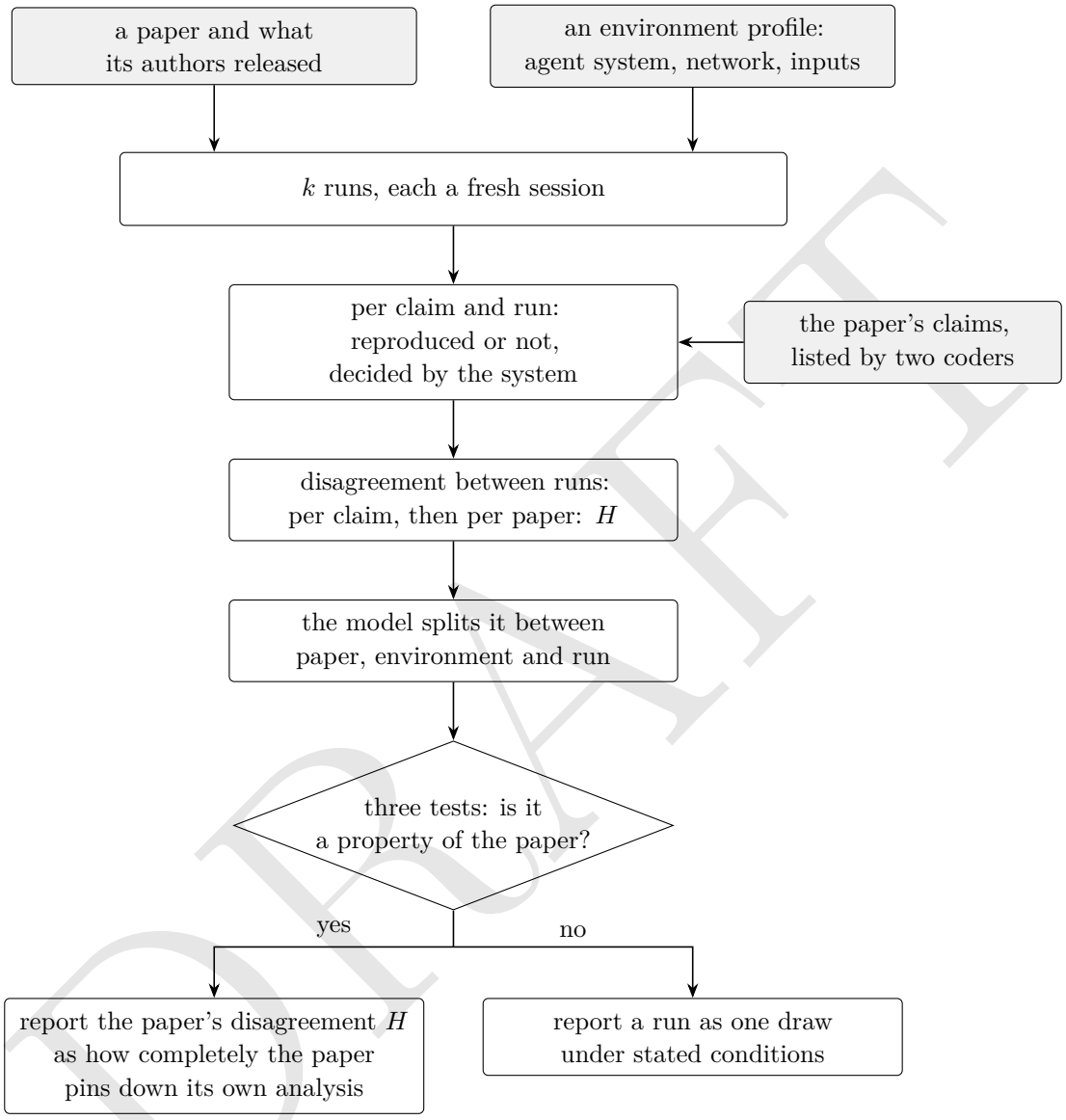


Figure 1: The study in one picture. H is the chance that two runs disagree about one result of the paper, averaged over its results. The shaded boxes are fixed before any run.

The evidence that prompted the question is a pilot of 49 runs of three papers (Appendix A): the spread between runs is real and large, and most of it in the pilot came from the agent system, the network and labeling rather than the paper. The model therefore treats the environment as a facet to estimate, not as noise, and gives each paper its own dispersion instead of pooling one.

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2 Setting

ReplicHAI audits the reproducibility of a research paper with a coding agent. Given a paper, one **run** finds what the authors released and splits the paper into components. It places each component in the inner loop, the part a machine can re-run (the analysis, the model or the pipeline), or in the outer loop, the study with human participants, which is never re-run or simulated. It re-runs the inner loop, marks each inner component verified, partial or blocked, records where its numbers disagree with the paper's, and lists the hidden decisions: choices a reimplementation has to make that the paper does not state. A second stage grades the run against a fixed rubric, and a run that falls short is revised.

Three things can differ between two runs of one paper, and the model keeps them apart:

- the **paper**, with the materials its authors released;
- the **environment** of the run: the agent system (the coding agent and the model behind it), the network it may reach, and the copy of the paper and of the artifacts it was given;
- the **run** itself: a fresh session of the same agent on the same inputs does not make the same choices twice.

A **claim** is one quantitative result the paper reports, such as a test statistic, an estimate or a count. Whether a run reproduced a claim is the unit the model is built on.

3 Why not plain generalizability theory

A generalizability study over paper × prompt × run, where the prompt is the instructions the agent is given and runs are nested in each paper and prompt cell, estimates one run variance component for the whole corpus, $\sigma^2_{r:pq}$, and treats it as error around each paper's mean. That is the right tool for asking how reliably a paper's *mean* outcome is measured. It cannot say that one paper's runs disagree more than another's, because it assumes they all disagree by the same amount. The question here is the opposite one, so the run variance has to be allowed to differ by paper and to be modeled in its own right.

4 Data and notation

Symbol	Meaning
$p = 1, \dots, P$	papers
$c = 1, \dots, m_p$	claims of paper p: the paper's reported quantitative results, frozen per paper by two human coders before any run is scored
$e = 1, \dots, E$	environment profiles: the agent system and model, the network policy, the copy of the paper, the artifact snapshot. A small number of designed levels, crossed with papers
$r = 1, \dots, k$	runs, nested in paper × profile
$Y_{pcer} \in \{0, 1\}$	whether claim c reproduced in run r: the run's observed value is within the registered tolerance of the reported value, decided by the system from the run's reported number, never by the agent's label

A claim the run did not attempt, or could not run, is $Y = 0$ and also carries a cause (environment, paper, artifacts) recorded by the run, so a blocked claim can be analyzed by why it was

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blocked.

5 Estimands

Reproduction probability of a claim. $q_{pce} = \Pr(Y_{pcer} = 1)$ for a fresh run under profile e.

Claim instability. The probability that two independent runs disagree about claim c:

$$h_{pce} = 2\, q_{pce}\, (1 - q_{pce}). \tag{1}$$

It is zero for a claim that always or never reproduces, and largest at $q = 1/2$.

Paper instability. The paper's disagreement H from the research question, the average over the paper's claims:

$$H_{pe} = \frac{1}{m_p} \sum_{c=1}^{m_p} h_{pce}. \tag{2}$$

This is the primary per-paper dispersion: the chance that two runs disagree about a claim of this paper, under this environment. It has a plain reading and a direct estimator, and it does not depend on how an agent decomposes the paper, because the claims are frozen.

Run-shared dispersion. How much a whole run moves together: a run that cannot reach the paper, or loses its dependencies, fails many claims at once. This is s_{pe} in the model below.

Environment effects. The shift in reproduction, β_e, and in run-shared dispersion, α_e, between profiles, and the paper × environment interaction v_{pe}: whether the environment changes *which* papers reproduce.

6 A model-free estimator

With $k \geq 2$ runs and x_{pce} successes on claim c,

$$\hat{h}_{pce} = \frac{2\, x_{pce}\, (k - x_{pce})}{k\, (k - 1)} \tag{3}$$

is unbiased for h_{pce}, and $\hat{H}_{pe}$ is the mean of $\hat{h}_{pce}$ over claims. Because the claims of a paper act as replicates, it is far more precise at a small k than a run-level standard deviation. In a simulation with 15 claims, a third of them unstable ($q = 0.5$) and the rest stable ($q = 0.95$), the 95% range of $\hat{H}/H$ was:

Runs per paper k	95% range of $\hat{H}/H$	Spread	Run-level SD, same data
3	0.39 to 1.55	4.0×	can be 0; unusable
5	0.64 to 1.39	2.2×	0.34 to 1.65
8	0.73 to 1.29	1.8×	0.49 to 1.51
10	0.77 to 1.26	1.6×	0.54 to 1.44

For comparison, the 95% interval on a single paper's run standard deviation from k normal draws is 0.52 to 6.28 times the estimate at $k = 3$, 0.60 to 2.87 at $k = 5$, and 0.69 to 1.83 at $k = 10$.

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7 The model

A claim-level logistic mixed model in which each paper has its own run-shared variance:

$$\operatorname{logit}\Pr(Y_{pcer}=1)=\beta_0+\beta_e+u_p+v_{pe}+w_{pc}+\eta_{per} \tag{4}$$

$$u_p\sim N(0,\tau_u^2),\qquad v_{pe}\sim N(0,\tau_v^2),\qquad w_{pc}\sim N(0,\tau_w^2) \tag{5}$$

$$\eta_{per}\sim N\big(0,\ s_{pe}^2\big)\,,\qquad \log s_{pe}=\alpha_0+\alpha_e+\omega_p,\qquad \omega_p\sim N(0,\tau_\omega^2) \tag{6}$$

$$\operatorname{corr}(u_p,\omega_p)=\rho. \tag{7}$$

- u_p is how reproducible the paper is overall; w_{pc} how hard each claim is.
- η_{per} is the run's shared shock: everything that moves all of a run's claims together. Its standard deviation s_{pe} belongs to the paper and the environment, which is the formal version of "each paper has its own dispersion".
- Claim-specific disagreement between runs, beyond the shared shock, is the Bernoulli variation of claims whose q is away from 0 and 1. That is where a decision the paper left open shows up: it moves one claim, not the whole run.

The same model as a picture. Each box on the left moves the chance that a claim reproduces. The box at the bottom sets how large a run's shock is, and it is where a paper gets a dispersion of its own:

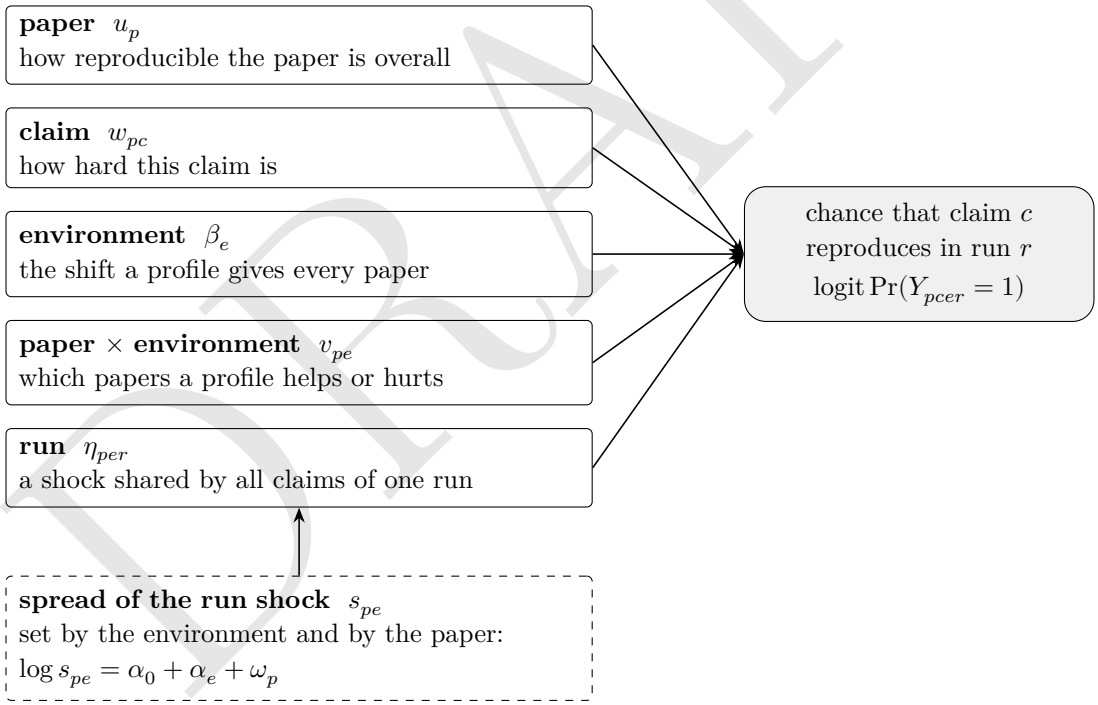


Figure 2: What moves a claim's outcome in the model. The dashed box is where a paper gets a dispersion of its own: ω_p is what the third research question asks about.

The split between the two is the one the pilot points at. On the latent scale, the share of a claim's between-run variance that is shared by the whole run is

$$\lambda_{pe}=\frac{s_{pe}^2}{s_{pe}^2+\pi^2/3}. \tag{8}$$

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A high λ says the runs differ as wholes: the environment, or the agent's trajectory. A low λ with a high H says individual claims flip independently: what the paper did not specify.

If a continuous run score is preferred to claim outcomes, the same idea is a location-scale model on the score S_{per} (for example the logit of the share of claims reproduced):

$$S_{per} = \beta_0 + \beta_e + u_p + v_{pe} + \varepsilon_{per}, \qquad \varepsilon_{per} \sim N(0, \sigma^2_{pe}), \qquad \log \sigma_{pe} = \alpha_0 + \alpha_e + \omega_p. \quad (9)$$

That version fits in brms as

```
bf(S ~ env + (1|i|paper) + (1|paper:env),
   sigma ~ env + (1|i|paper))
```

The claim-level version needs a short Stan program, because the run effect's standard deviation is itself hierarchical across papers. Both are fit on all runs at once, which handles unequal numbers of runs and claims.

8 Is dispersion a property of the paper?

Three tests, registered before any outcome is seen. All three have to pass for the study to say that dispersion describes the paper.

1. **Between-paper variance of dispersion is above zero.** $\tau^2_\omega > 0$ in the model, by a likelihood-ratio test against the model without ω_p (the null is on the boundary, so the reference distribution is a 50:50 mixture of χ^2_0 and χ^2_1), or by its posterior. For the model-free estimator, the between-paper variance of $\hat{H}_p$ must exceed the mean of its within-paper sampling variance, estimated by resampling runs.
2. **The ranking survives a change of environment.** Papers ranked by H under two agent systems must agree (Spearman correlation registered in advance). A property of the paper does not reorder when the agent changes. The pilot already fails this on four of six crude measures, with too few runs to decide.
3. **Dispersion follows what was withheld.** Take papers whose analysis is fully specified. Run them as published, then with one stated detail removed (a seed, a degrees-of-freedom method, an exclusion rule). The claims that detail affects must gain instability and the others must not. The unedited papers also give the instrument's noise floor: their H is what the agent and environment contribute alone.

9 Reliability

For ranking papers by reproduction level, the generalizability coefficient with paper as the object of measurement is

$$E\rho^2 = \frac{\tau^2_u}{\tau^2_u + \tau^2_v/n_e + \sigma^2_{\text{res}}/(n_e\, k)}. \quad (10)$$

For dispersion, the reliability of $\hat{H}_p$ is the share of its variance across papers that is not sampling noise. Estimate it by splitting each paper's runs into halves, correlating $\hat{H}_p$ across papers between the halves, and applying the Spearman-Brown correction. Require at least 0.8 before reporting a single paper's dispersion.

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A criterion proposed earlier in this project, "if prompt-level variance exceeds run-level variance we do not have an instrument", compares two error terms. A prompt's main effect moves every paper equally and does not change a ranking; what matters is the paper × prompt interaction relative to τ_u^2, which $E\rho^2$ already contains. Three prompts give two degrees of freedom for a prompt main effect, so prompts are a fixed facet, and no claim generalizes beyond the prompts tested.

10 Sample size

The intended corpus is the 76 CHI papers that have shared both data and code, whose reproduction outcomes were verified by hand (Iarygina, Hornbæk and Mottelson, 2026). That caps the corpus at 76 papers, so runs per paper is the lever that is left. Pooled components are well estimated at $k = 3$; per-paper dispersion is not. A design that can grow into the main study:

Stage	Papers × runs	Purpose
Internal pilot	16 × 8, one profile	Estimate τ_ω^2 and the reliability of $\hat{H}_p$. These runs count toward the main study
Main study	the other 60 × 5 if τ_ω^2 is detectable, otherwise × 3 and no per-paper dispersion claims	Paper levels and, if supported, dispersion
Environment facet	20 × 2 more profiles × 4	β_e, α_e, the paper × environment interaction, and test 2
Noise floor and withholding	5 to 8 fully specified papers, as published and with one detail withheld	Test 3

The decision rule between "× 5" and "× 3" is written into the registration.

11 Assumptions and open questions

- Runs are independent given paper and profile. This needs fresh sessions with no shared cache, recorded inputs, and the analysis of the first graded attempt, before any revision the grader requests.
- Claims are fixed by people, with agreement between two coders reported. Claims that are qualitative are out of scope and counted, not dropped.
- The match tolerance is registered per claim type before runs are scored.
- How launch failures are excluded is stated before outcomes are seen.
- A run's blocker cause is recorded in a closed list, so environment-caused zeros can be separated from paper-caused zeros in a sensitivity analysis.

References

Iarygina, O., Hornbæk, K., and Mottelson, A. (2026). On the Computational Reproducibility of Human-Computer Interaction. *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, 1–16. https://doi.org/10.1145/3772318.3791129

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A What repeat runs of one paper showed

Three papers were audited ten to twenty-one times each on 10 and 11 September 2026, all under one method, meaning one set of instructions and one rubric (digest `sha256:f9af95bb…`). This appendix records what those runs show about the idea that run-to-run variation measures what a paper left unsaid. The data was analyzed on 9 October 2026.

The short answer: the runs vary a lot, but most of the variation traced so far comes from the agent system, the network, how the agent labels its own results, and the grader's revisions. The paper's own silence is a real part of it, and a small one. A design that reads the spread as a property of the paper has to remove or measure those other sources first. The main text proposes the measurement.

A.1 The runs

Paper	Called below	Runs started	With a verdict
Johnson, Sengupta, Schöning and Hecht (2016). *The Geography and Importance of Localness in Geotagged Social Media.* CHI '16, 515–526. 10.1145/2858036.2858122	Geography	10	10
Ou (2026). *Progressive Autonomy as Preference Learning: A Formalization of Trust Calibration for Agentic Tool Use.* arXiv:2605.19151	Progressive autonomy	18	6
Kay, Matuszek and Munson (2015). *Unequal Representation and Gender Stereotypes in Image Search Results for Occupations.* CHI '15, 3819–3828. 10.1145/2702123.2702520	Unequal representation	21	10

Each run is its own private repository under `open-hai`, named after the paper with a four-character suffix. The suffixes below refer to those repositories.

The runs with no verdict failed at launch. Eight repositories on the third paper were created within 20 seconds of each other and hold no commits. Most of the rest hold only the system's identity commit, written seconds after the repository was created, with no agent commit after it. Runs started later the same day completed 16 of 16. The completion rates in the table therefore say when each paper was run, not how hard it is. No published record says why a run stopped, so this reading rests on timing, not on a recorded cause.

A.2 Two agent systems ran every paper

Each run's `environment.json` names its harness and model. The pilot mixed Claude Code on `claude-opus-5` with Codex on `gpt-6-astra`, under the same method digest. The manifest the registry reads, `replication.json`, records neither, so the mix is invisible to anything that

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reads only the manifest.

Paper	System	Runs	Inner components	Verified rate, mean and SD	Hidden decisions	Outer-loop rows
Geography	Claude Code	5	14 to 22	38%, SD 18 pp	12 to 22	2 to 5
Geography	Codex	5	18 to 21	28%, SD 5 pp	17 to 28	2 to 4
Progressive autonomy	Claude Code	3	14 to 21	61%, SD 10 pp	10 to 22	4 to 5
Progressive autonomy	Codex	3	16 to 18	67%, SD 14 pp	21	1
Unequal representation	Claude Code	5	11 to 16	73%, SD 9 pp	13 to 22	5 to 6
Unequal representation	Codex	4	12 to 15	61%, SD 9 pp	17 to 24	5

The verified rate is the share of a run's inner-loop components that the run marked verified. One run on the third paper left no `environment.json` and is not in this table.

- **Spread depends on the system.** Within one paper, the component count varies about eight times more under Claude Code than under Codex (mean variance 10.6 against 1.3).
- **So does the hidden-decision count.** Codex lists four to five more hidden decisions per run on the same paper. The paper means barely differ (17.8 to 19.4), so the count tracks the agent, not the paper.
- **So does the boundary.** On progressive autonomy, Codex puts one component in the outer loop and Claude Code puts four or five. The two systems slice the paper differently: Codex lists pipeline modules, Claude Code lists claims.
- **The ranking of papers by spread is not stable across systems.** A review pass found that the order of the three papers flips between systems on four of six measures. With three to five runs per cell this is suggestive, not established.

A.3 The same number, read and labeled differently

On *Unequal representation*, the runs mostly computed the same values. What differed was the paper they compared against and the label they gave the result.

- **Which copy of the paper a run could read.** The Claude Code runs were refused by the sandbox's network allowlist when they tried the publisher and the authors' pages (m9kk records "NOT FETCHED - the allowlist refused both domains"; 5nbs and k38p record the same). They took reference values from the authors' rendered notebook instead. The Codex run aysp fetched an author PDF through its web reader. The two copies do not agree in every number, so the same computation was compared against different reported values.
- **Identical disagreements, different labels.** Three runs (k38p, qzfk, r82q) recorded the same disagreement on the "trustworthy" coefficient in the paper's Table 1: reported -0.1117, observed -0.0463. Two labeled the claim *partial*, one *verified*.
- **Factual items answered differently.** Whether the paper's data was released is a fact, yet on progressive autonomy three of six runs say yes and three say no. Two of them

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(c9ks and k46m) describe the same release and still answer differently.

A difference of this kind is not underspecification. It would disappear if every run read the same copy of the paper and the system, not the agent, decided whether a number matched.

A.4 The grader changes what is measured

Every run is graded inside the loop, and a run that falls short is revised. In four of the six runs where a revision happened and the history kept both versions, the revision changed fields the analysis would measure:

- *released* rows grew from 23 to 30 (2mzg), 29 to 30 (grfb) and 36 to 37 (akwe);
- one run's component count went from 11 to 12 and its decisions from 16 to 17 (kpam);
- one run's mismatch count went from 6 to 8 (akwe).

A review pass found that Codex runs took two grading rounds in 10 of 12 cases and Claude Code runs one round in 10 of 13, so the push toward the rubric is itself system-dependent. The grader always runs on the worker's model.

A.5 The findings file was often unreadable to the system

The system rejected 14 of the 27 findings files it read when finalizing the pilot runs: all 12 from Codex and 2 of the 12 from Claude Code. The agents wrote mismatch values as objects and numbers, or a fetch the sandbox refused as `not_fetched`, and the schema allowed neither. One mistyped field discarded the whole file, with a message saying the file was not valid JSON, which was false. Every file was valid JSON. The model grader passed the schema criterion on all 14.

Schema 2 of `verdict.json` reads what the agents wrote. Run over the 29 findings files the pilot runs published, it accepts 27 and reads all 214 of their mismatches. The two it still refuses break a real rule, a search recorded as found with no address, and each is reported at its row with the rest of the file kept.

A.6 Freezing the denominator does not remove the swing

An earlier plan proposed fixing the denominator of the verified rate, because the agent picks its own component count. Holding each paper's denominator at its mean leaves the spread as large as before (SD 17, 10 and 12 points across the three papers, against 14, 11 and 10 observed). The numerator moves as much as the denominator.

The runs already extract the paper's own headline results, in `instrument.json`. Those lists have a stable core and a long tail:

Paper	Runs with a list	Entries per run	Distinct numbers	In every run	In at least half	In one run only
Geography	7 of 10	9 to 16	83	10	36	21
Progressive autonomy	0 of 6	none	none			
Unequal representation	10 of 10	10 to 18	94	3	36	24

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A claim list taken from one extraction would freeze one arbitrary draw. And *Progressive autonomy* reports simulation means and formal statements rather than tests, so a list limited to "a test carrying a number" would be empty for it.

A.7 What is the paper's own silence

A review pass reading the runs side by side found real gaps the paper leaves open: the tail of a Vuong test, a bootstrap seed, whether the first or the last selected image was used (5.307 against 5.300), and one contrast. These are the target of the measurement. In this pilot they are a small share of what moved between runs.

A.8 What this changes

- One run is one draw. A verdict from a single run should say so.
- Record every facet that can vary between runs in the manifest: system, model, network policy, paper copy, grading rounds, and why a run stopped.
- Give every run the same inputs, and let the system decide whether a number matched.
- Measure dispersion per claim, against a claim list coded by people.
- Treat the environment as a facet to record and vary on purpose, because it moves the result as much as anything else measured here.

A.9 How the numbers were checked

The tables in this appendix were computed from the records each run published, by a script kept with the ReplicHAI source.

The fetch-refusal quotes, the labels on that coefficient, and the before-and-after revision counts were read from the files directly. Three statements come from independent review passes and were not recomputed: the ranking flips between systems, the grading-round counts by system, and the list of genuine gaps.

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